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A genetic algorithm with a new local search method for solving the multimode resource-constrained project scheduling problem

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ABSTRACT

The existence of limited resources and precedence relations between activities makes project scheduling a difficult task. In this paper, the multimode resource-constrained project scheduling problem (MRCPSP) as one of the important project scheduling problems is focused. In this regard, a genetic algorithm (GA) is proposed for solving the multimode resource-constrained project scheduling problem (MRCPSP) improved by a new local search method. The Activity Mode List (AML) is used as encoding and the multimode serial schedule generation scheme (MSSGS) is employed as the decoding procedure. The aim of the suggested local search is maximum use of existed nonrenewable resources of the project such that its makespan is decreased. For this purpose, the local search procedure allocates the unused nonrenewable resources of each feasible solution to the activities with lower *total float*. Comparing results of the proposed method with other ones using the set J20 in Project Scheduling Problem Library (PSPLIB) validates the effectiveness of the proposed algorithm to solve the MRCPSP.

KEYWORDS

Genetic algorithm; multimode scheduling; resource constraints; local search strategy

Introduction

Multimode Resource Constrained Project Scheduling Problem (MRCPSP), as an extension version of RCPSP, has been attended very much by researchers in recent years. Under the resource and precedence constraints, the aim of MRCPSP is to assign an execution mode and a feasible start time for each activity, such that makespan of the project is minimized (Hartmann and Briskorn 2010; Chaleshtari and Shadokh 2014; Qi et al. 2014; Ahmed et al. 2018; Chakraborty et al. 2018; Muritiba et al. 2018; Almeida et al. 2019).

There are three procedures for solving the MRCPSP: the exact methods, heuristic and meta-heuristic approaches. Sprecher and Drexel (1998) presented that the exact procedures cannot solve the problems with more than 20 activities in an acceptable computational time. On the other hand, many researchers showed that the meta-heuristic approaches outperform the heuristic ones (Sebt et al. 2017). Thus, the meta-heuristic approaches are the best existing methods for solving the MRCPSP.

Some of the best meta-heuristic approaches applied for solving the MRCPSP are the following ones: Józefowska et al. (2001) applied a simulated

annealing (SA) algorithm with and without a penalty function and presented that the fitness function with penalty function has a better performance. Lova et al. (2009) proposed an efficient hybrid genetic algorithm for solving the MRCPSP, in which, they proposed a procedure to maximize the probability of obtaining feasible solutions in the initial population. Peteghem and Vanhoucke (2009) employed the Artificial Immune System (AIS) to solve the non-preemptive MRCPSP. Peteghem and Vanhoucke (2010) also suggested a genetic algorithm for the preemptive and nonpreemptive multimode RCPSP and evaluated the impact of preemption on the quality of the schedule. A hybrid rank-based evolutionary algorithm was applied for multimode RCPSP by Elloumi and Fortemps (2010) in which the problem of single objective MRCPSP was transformed to a bi-objective one to cope with the violation of nonrenewable resource constraints. Wang and Fang (2011) proposed an effective Shuffled Frog-Leaping Algorithm (SFLA), in which a new encoding scheme was adopted and a set of suitable parameters of the SFLA was determined using the Taguchi method of the Design of Experiment (DOE). Wang and Fang (2012) also proposed an effective Estimation of Distribution Algorithm (EDA) for the

MRCPSP. In the EDA, the impact of population size on the quality of the solutions was analyzed.

In the present study, GA as a general search strategy is proposed for solving the MRCPSP improved by a new local search strategy. The reason of the employing the local search strategy is maximum use of existed nonrenewable resources of the project such that its makespan is reduced. For this purpose, this strategy allocates the unused nonrenewable resources of each feasible solution to the activities with low *total float*. The local search is applied in the multi-mode forward-backward iteration (MM-FBI).

The rest of this paper is organized as follows: In Problem description section, the MRCPSP is described; Genetic algorithm section deals with the definition of GA for solving the MRCPCP; in Computational experiments section, the results of computational experiments to prove the effectiveness of our method in solving the MRCPSP are reported and finally, in Conclusion section, concluding remarks are drawn out.

Problem description

Generally, the MRCPSP can be stated as follows: A project has $J+2$ activities that each one can be performed in one out of the several modes with specific duration and resource requirements. An activity j ($j=1, \dots, J+1$) that started in mode m ($m_1, \dots, M_j$) must be completed in mode m without preemption (Peteghem and Vanhoucke 2009). The start and end activities of the project are dummy activities with zero duration and no resource requirements. These Dummy activities are 0th and $J+1$ th activities, respectively. There are precedence relations between some of the activities. In this paper, these precedence relations are finish-start relations without time lags. Two categories of resources are considered for each project: renewable and nonrenewable resources. In each project, the availability of renewable resources k ($k=1, \dots, R$) and the availability of nonrenewable resources l ($l=1, \dots, N$) are considered R_k and N_l , respectively. If activity j ($j=0, \dots, J+1$) is performed with mode m , we present its duration, its required amount of renewable resources k ($k=1, \dots, R$) and its required amount of nonrenewable resources l ($l=1, \dots, N$) with d_{mj} , r_{jmk} , and n_{jml} , respectively. Under the above mentioned conditions, the aim of the MRCPSP is to find a feasible start time and a mode to each activity such that makespan of the project is minimized.

Genetic algorithm

John Holland developed and introduced Genetic Algorithms (GA) in the 1960s and 1970s. The bases of GA are the mechanics of natural selection and natural genetics (Damak et al. 2009). It is established via a collection of preliminary feasible solutions that are randomly generated or created using a heuristic entitled 'the initial population'. Every member of the so-called population is known as a Chromosome that indicates a solution to the problem. Chromosomes comprise a sequence of elements entitled 'genes'. GA requires a suitable encoding for each problem. Furthermore, it needs a fitness function that acts as a measure of the quality of every existing chromosome in the population. In this method, the reproduction tool attempts to choose the parents via a crossover operator that generates offspring chromosomes. These offspring are capable of randomly undergoing a mutation operation by locally changing the genes, and the elitism mechanism is used to copy the best chromosomes to the new population. Upon creation of the new population, the resulting chromosome elements are assessed based on their specific fitness values. This iterative process continues until the termination condition is achieved (Fung et al. 2008; Sarkar 2018).

A genetic algorithm for solving the MRCPSP

The proposed GA for solving the MRCPSP has been depicted in Figure 1. In following, all steps of this procedure will be explained.

Encoding scheme

Selecting a suitable representation is essential for the search (Lee and El-Sharkawi 2008). In this study, the Activity Mode List (AML) is used for representing chromosomes. AML comprises two sections: an Activity List (AL), $\lambda = \{A_0, A_1, \dots, A_{J+1}\}$, and a Mode assignment List (ML), $\mu = \{M_0, M_1, \dots, M_{J+1}\}$. M_i in the ML specifies the mode of A_i in the AL. Hartmann and Kolisch (2000) established through experiments that the procedures based on AL representations overtook other procedures. The mode assignment list in the MRCPSP is required to allocate a mode to each activity. Hence, in this study, the AML is reflected for chromosome representations.

Initial population

The initial population of the study can be produced randomly or via heuristic methods. In this study, the

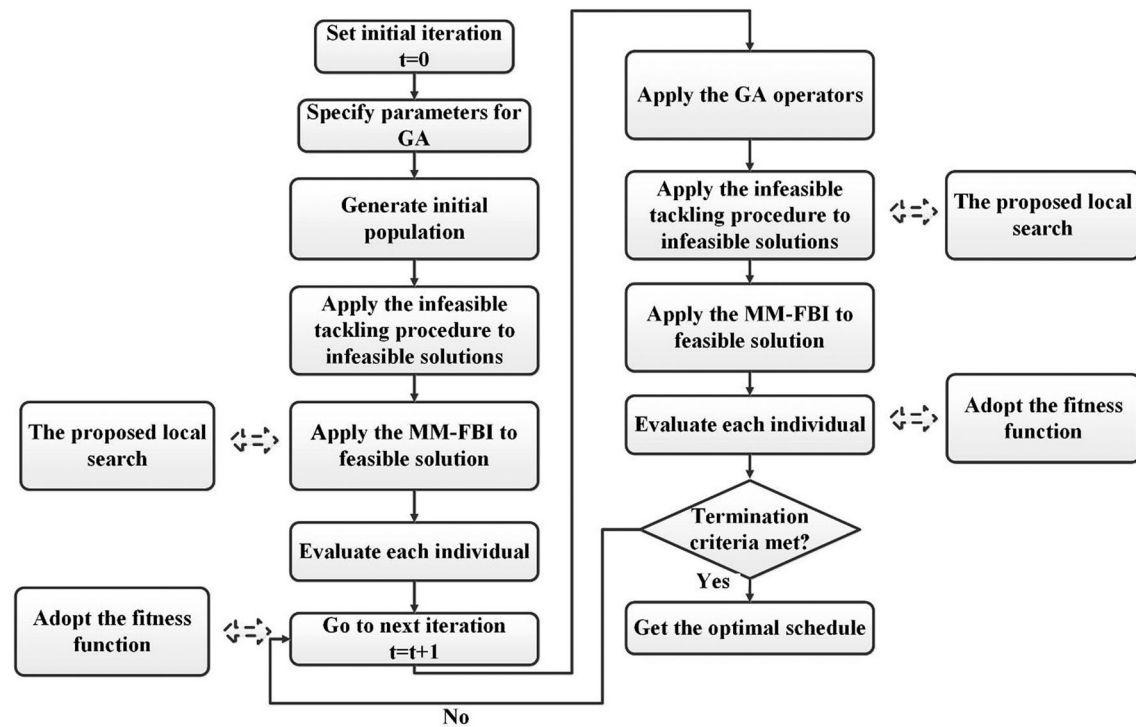


Figure 1. The procedure of the proposed GA for solving the MRCPSp.

initial population is established randomly to diversify the search space with the purpose of obtaining favorable regions (Lee and El-Sharkawi 2008).

Decoding procedure (schedule generation scheme)

The role of the schedule generation scheme (SGS) is to convert a representation solution of the RCPSP into a schedule (Lova et al. 2009). Two types of SGS exist, namely serial SGS and parallel SGS. Considering the parallel SGS lacks the ability to obtain the optimal solution every time (Lova et al. 2009), thus, the serial SGS is utilized. In the MRCPSp, the serial SGS needs to be adjusted to address nonrenewable resources issues. In this course, two essential matters should be reflected: first, transforming infeasible solutions to feasible ones as much as possible (i.e. the repair strategy), and second, fully exploiting nonrenewable resources within the feasible solutions. In this sense, both the infeasible and feasible procedures are employed for this purpose. It is noteworthy to say that these procedures are an extension of the serial SGS, which is known as the multimode serial schedule generation scheme (MSSGS).

Infeasible tackling procedure (repair strategy)

The best method for tackling infeasible solutions is to apply the repairing procedure (Lee and El-Sharkawi

2008). Therefore, in this study, once the initial population is generated, the infeasible tackling procedure is used to transform infeasible solutions into feasible ones. In this course, first, an activity is chosen randomly whereby its mode is modified resulting in a new mode. On the condition that ERR (μ) is less than its predecessor is, the mode is altered by replacement with a new mode. This procedure persists until all the infeasible solutions transform into feasible ones, or until the predetermined maximum number of iterations for SGS (SGS_adj) equals the number of non-dummy activities (J). Following the procedure, the feasible solutions are filtered and allowed entry to the multimode forward-backward iteration (MM-FBI) method, which is defined in the following. The feasible tackling procedure is applied in the MM-FBI.

Multimode forward-backward iteration (MM-FBI) method (local search strategy)

In the RCPSP, the Forward-Backward Iteration (FBI) is an effective suitable method for improving the solutions' quality (Lova et al. 2009). Lova et al. (2009) further developed the idea for the MRCPSp (known as the multimode forward-backward iteration (MM-FBI) method). This method decreases the completion time of every activity separately at every decision point of the partial schedule by modifying its mode without altering the modes and deferring the

completion times of other activities. Yet, only some activities are chosen in this regard. Now this question arises ‘which activities should be applied to the mode improvement procedure?’

The local search proposed by Lova et al. (2009) is based on the random method (i.e. for this purpose, related activities are selected randomly). Nonetheless, the local search method by Sebt et al. (2017) is based on an algorithm in which activities are selected for the mode improvement procedure that by changing their mode, their renewable resources and their durations reduce more than other activities.

In this study, other local search methods are presented. We believe that the activities possessing a lower *total float* are potentially more efficient in the reduction of makespan.

It has long been observed that the reduced duration of critical tasks (i.e. activities with zero total float) reduces the project’s duration, provided unlimited quantities of resources are available (Jiang and Zhu 2010; Lu et al. 2017). Consequently, in MRCPSp, it is predicted that if the mode improvement procedure is implemented on tasks with a low *total float*, thus, makespan will reduce more efficiently. To achieve this result, we suggest that in each backward scheduling of each solution, the activities with a less *total float* are chosen for the mode improvement procedure. This action is also taken in each forward scheduling. The flowchart of the proposed method is depicted in Figure 2.

SGS_Pper states the percentage of total nondummy activities chosen for the mode improvement procedure. The MM-FBI repeats until no additional enhancement of the makespan is possible.

Illustrative example for demonstrating the effectiveness of the proposed local search

To compare the performance of the proposed local search with Sebt et al. (2017)’s one, the example used in Sebt et al. (2017) is solved.

The mentioned example includes theoretical activities. The number of activities in the example has been kept small in order to enhance the clarity of presentation in the suitable tables and figures. However, the data concerning the modes of activities, their renewable and nonrenewable resources are sufficiently diverse to illustrate the computations involved, and demonstrate the essential features of the proposed method. The example includes nine non-dummy activities, two dummy activities, one renewable resource type, and one nonrenewable resource type. Here, each activity has two modes. Start and end

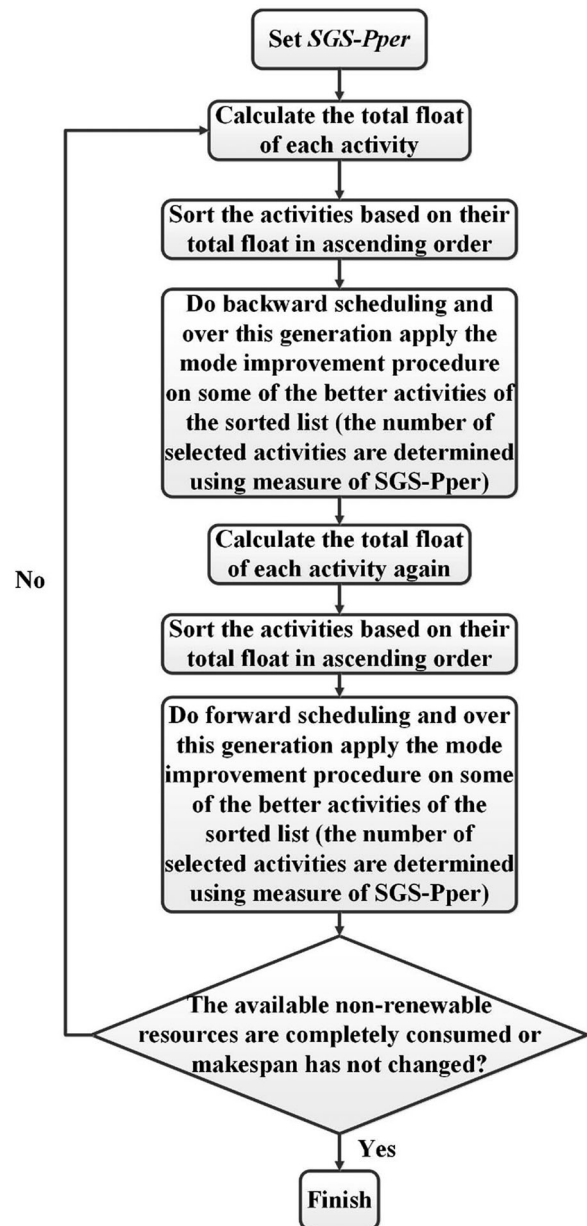


Figure 2. The proposed local search applied in the MM-FBI.

activities are dummy activities with merely one mode, zero durations, and zero consumption of resources. The availability of renewable and nonrenewable resources is 2 and 28, respectively. Detailed information and precedence constraints of this project are given in Table 1 and Figure 3, respectively.

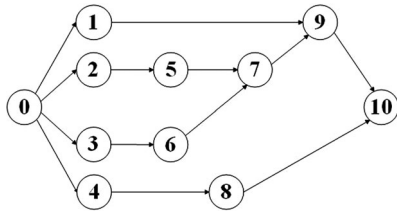
In this example, consider solution A which is given in Figure 4.

A resource graph of this solution (Figure 5(a)) was developed using the SSGS procedure (3.1 means activity 3 with mode 1).

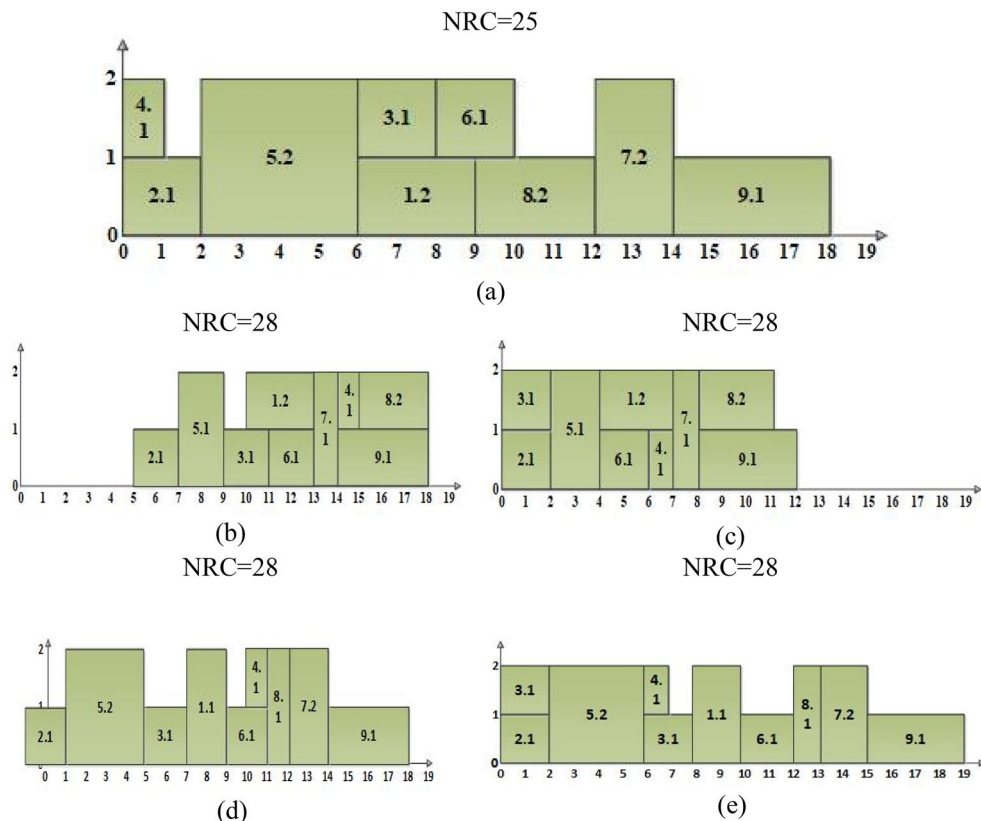
The MM-FBI is established with backward scheduling on the existing schedule. For this purpose, the completion time of activities in the present schedule

Table 1. Data for activities in the example project.

Activity no.	Mode 1			Mode 2		
	r_{j1k}	n_{j1l}	d_{1j}	r_{j2k}	n_{j2l}	d_{2j}
0	0	0	0	0	0	0
1	2	4	2	1	2	3
2	1	5	2	1	2	4
3	1	3	2	1	1	3
4	1	2	1	1	1	2
5	2	5	2	2	3	4
6	1	4	2	1	3	3
7	2	2	1	2	1	2
8	2	3	1	1	2	3
9	1	3	4	1	1	6
10	0	0	0	0	0	0

**Figure 3.** Activity network of an example of the MRCPSp.

Solution A	$\lambda =$	0	2	4	5	1	3	6	8	7	9	10
	$\mu =$	1	1	1	2	2	1	1	2	2	1	1

Figure 4. An example of a particle representing a solution for the MRCPSp.**Figure 5.** A single iteration step of the MM-FBI and nonrenewable resource consumed (NRC). (a) Existing schedule, (b) backward scheduling using proposed local search or local search of Sebt et al. (2017), (c) forward scheduling using proposed local search or local search of Sebt et al. (2017), (d) backward scheduling using Lova et al. (2009) local search, (e) forward scheduling using Lova et al. (2009) local search.

(Figure 5(a)) is taken as a measure of activity priorities while the mode improvement procedure is executed over the generation of the schedule on activities possessing less *total float*. For Solution A, total float of each activity is given in Table 2.

According to this table, it is shown that Activity 2, Activity 5, Activity 7, and Activity 9 are the critical tasks (i.e. their total float equals zero). Consequently, the selected activities are chosen for the mode improvement procedure in backward scheduling. Figure 5(b) shows that the mode improvement procedure reduces the duration of Activity 5 by two periods and lowers the duration of Activity 7 by one period.

According to the previously mentioned changes, consumption of the nonrenewable resource is adjusted from 25 to 28 (i.e. the full capacity of the nonrenewable resource is used). While a single iteration of the MM-FBI is completed with the forward

Table 2. Total float of each activity for solution A.

Activities	1	2	3	4	5	6	7	8	9
Total float	5	0	2	8	0	2	0	8	0

scheduling, start times of activities in the existing backward schedule are assigned for activity priorities. Meanwhile, the available nonrenewable resources are depleted; thus, the mode improvement procedure is not implemented in this direction.

After completing the single iteration of the MM-FBI, the makespan of the project reduces from 18 to 12 (see Figure 5(c)) while available nonrenewable resources are fully consumed ($NRC = 28$). Consequently, the MM-FBI method is not applied again.

On the other hand, we apply the forward-backward scheduling using the random method. In this method, more activities compared to the proposed method are tested, since high potential activities are not known. To achieve this, a larger value should be considered for SGS_Pper . Wang and Fang (2011) specified that SGS_Pper is the most important factor in the solutions' quality. They (2011) also revealed that the best value for SGS_Pper in the random method is 0.7. Nonetheless, in our proposed method, it is safe to consider SGS_Pper equal to 0.22 ($2/9 = 0.22$, 2 is the number of selected activities for the mode improvement procedure, and 9 is the number of non-dummy activities). While the schedule developed using the MSSGS is calculated based on Equation (7), therefore, using the proposed method, more evolution is observed in the population; thus, the algorithm may converge to a better solution.

$$\frac{\sum_{j=1}^J (SGS_{Pper} * M_j + (1 - SGS_{Pper}))}{J} = 1 + 2 * SGS_{Pper} \quad (1)$$

When the random method is implemented in Solution A, it is possible to select the activities 1 and 8 to improve the makespan. As shown in Figure 5(d,e), while the duration of activities 1 and 8 are reduced, makespan of the project has increased. In addition, if the local search method proposed by Sebt et al. (2017) is applied to Solution A, similar activities with the proposed local search are selected; hence, makespan of the project is reduced from 18 to 12 (see Sebt et al. 2017).

Comparing the makespan of our local search procedure (i.e. 15) with the makespan reported by Lova et al. (i.e. 17), we observe that our proposed procedure acts more efficiently in reducing the makespan of the project. In addition, the proposed local search

method offers similar performance to that of the local search method offered by Sebt et al. (2017).

In order to prove the effectiveness of the proposed local search procedure, the suggested method and Sebt et al. (2017) one were applied on the well-known benchmark set of J20 generated by the project generator ProGen for the MRCPSP. The results showed that both of methods have approximately equal performance.

Fitness function

Determining a suitable fitness function is essential for the accurate operation of evolutionary algorithms. In every generation, all chromosomes in the population are assessed based on the selected fitness function. The chromosomes with better fitness values are entered in the mating pool to produce new offspring. Therefore, selecting chromosomes that possess better fitness values allows the GA to obtain solutions within a shorter period.

Numerous fitness functions have been introduced in the MRCPSP literature. Considering the superiority of the fitness function proposed by Sebt et al. (2015) (Equation 3), we have used the same in the present study.

$$f(I) = \begin{cases} 1 - \frac{T - \text{mak}(I)}{T}, & \text{if } \text{ERR}(\mu) = 0 \\ 1 + \frac{\text{mak}(I)}{T} + \text{ERR}(\mu), & \text{otherwise} \end{cases} \quad (2)$$

$$\text{ERR}(\mu) = \sum_{l=1}^L \max \left\{ 0, \frac{\sum_{j=1}^J n_{jml} - N_l}{N_l} \right\} \quad (3)$$

In this fitness function (Equation (5)), $\text{ERR}(\mu)$ penalizes the infeasible solutions and demonstrates the nonrenewable infeasible degree of ML (Equation (6)). Evidently, when $\text{ERR}(\mu)$ is zero, the individual I is considered a feasible solution. The other parameters of this fitness function are as follows:

T gives the upper bound on the project's makespan determined by the sum of the maximal duration of activities and $\text{mak}(I)$ is the makespan of the individual I .

Selection

The selection operator utilizes available fitness information to favor higher quality individuals so that their information is transferred to the next generation of the evaluation. It provides a mechanism to affect this balance towards an exploitation by creating

emphasis on the better individuals or towards exploration—by providing similar chances to survive even for worse individuals’ (Lee and El-Sharkawi 2008). In order to balance between exploration and exploitation, the present study randomly selects the father and mother from the best individuals’ pool (20% of top individuals) and total individuals, respectively.

Crossover operator

The crossover operator is executed on each pair of parents, which produces two children that inherit sections of their parents’ characteristics. When using a crossover, both AL and ML are adapted. Peteghem and Vanhoucke (2010) inspected various potential crossover operators and verified that the one-point crossover offers the best performance in the MRCPSp. Thus, the one-point crossover is employed in this study. ‘In the one-point crossover, an integer number r is randomly selected. All data left from that point, both from the activity list and from the mode list, is copied from the father’s chromosome’.

Mutation operator. A mutation operator is a vital tool for the maintaining diversity inside the population and to avoid premature convergence (Lee and El-Sharkawi 2008). Sebt et al. (2017) presented a new mutation operator for AL and ML, which offered favorable results for the MRCPSp. Hence, in this study, it is employed as follows:

In AL, numerous activities in diverse or similar solutions are randomly selected while their priorities are also altered randomly. In ML, the mutation operator performs differently based on whether the solution is feasible or infeasible. For a feasible solution and a randomly selected activity, provided the new mode list remains feasible and $\eta(\mu)$ in Equation (4) rises, a new mode replaces the preceding mode of that activity; else, nothing is modified. Evidently, through Equation 4, $\eta(\mu)$ for the feasible solution is assigned a negative or zero value; it exploits nonrenewable resources.

$$\eta(\mu) = \sum_{l=1}^N \frac{\sum_{j=1}^j n_{jml} - N_l}{N_l} \quad (4)$$

On the other hand, for the infeasible solution and a randomly selected activity, on the condition that the ERR (μ) of the new mode list decreases, the previous mode of the activity replaces the new one; else, nothing is modified. The proposed mutation operator repeats until the stated mutation rate is achieved.

Computational experiments

In this section, the proposed method is examined in solving the MRCPSp. The offered GA was prepared and tested using MATLAB R2014b software on a laptop with an Intel® Core 2 T9300 2.5 GHz processor.

In similar studies, when comparing various algorithms, the CPU time and the number of generated schedules are determined as the stopping criteria. Since 5000 generated schedules halting criterion has been used more than other halting criteria by the researchers for the comparison, it is also used in the current study.

To elucidate the effectiveness of the proposed GA, the famous benchmark set of J20 generated using ProGen for the MRCPSp was used (available at <http://129.187.106.231/psplib/>). The set comprises 640 instances with 20 nondummy activities. In every instance, an activity can be accomplished via any of the three modes with durations ranging from 1 to 10 periods. Each instance encompasses two renewable types of resources and two nonrenewable types.

Comparison with existing algorithms

Average deviation from the optimal makespans and the average CPU time in seconds are used for comparison. Optimal values for sets of 20 are considered as the base for calculating the average deviations. Accordingly, our results are compared with the existing state-of-the-art methods for solving the MRCPSp such as the Modified Variable Neighborhood Search Heuristic proposed by Chakraborty et al. (2019) denoted as CMVNSH; the GA and Hybrid GA-FIPS suggested by Sebt et al. (2015; 2017) denoted respectively as SGA and SHGF; GA and the AIS presented by Peteghem and Vanhoucke (2009, 2010) denoted respectively as VPVGA and VPVAIS, EDA and SFLA presented by Wang and Fang (2011, 2012) denoted respectively as LCEDA and LCSFLA, the hybrid genetic algorithm developed by Lova et al. (2009) denoted as LHGA, the hybrid rank-based evolutionary algorithm proposed by Elloumi and Fortemps (2010) denoted as EFEA, controlling search using an S decreasing constriction factor suggested by Chen and Wang (2013) denoted as CWC, discrete particle swarm optimization presented by Cui and Yu (2014) denoted as CYDP, hybrid estimation of distribution algorithm with random walk local search proposed by Soliman and Elgendi (2014) denoted as SEEDA, ant colony optimization developed by Li and Zhang (2013) denoted as LZA, cooperative discrete particle swarms suggested by Shen and Li (2013) denoted as

Table 3. Average deviations from optimal makespans for J20 (5000 generated schedules as the stopping condition).

Algorithm	Average deviations	Algorithm	Average deviations
CMVNSH	0.27	LZA	1.1
SHGF	0.41	JRPLA	1.24
SGA	0.56	LCEDA	1.28
VPVGA	0.57	LCSFLA	1.40
Proposed GA	0.68	SLC	1.62
VPVAIS	0.70	JRR	1.55
CWC	0.71	EFEA	1.62
LHGA	0.87	ZLZH	1.82
CYDP	0.97	CLPE	5.78
SEEDA	1.09		

Table 4. Comparison with other proposed algorithms considering average CPU time for J20 (5000 generated schedules as the stopping condition).

Algorithm	Proposed GA ^a	SGA ^a	SHGF ^a	VPVGA ^b	LHGA ^c	LCSFLA ^d	JRR ^a
J20	0.17	0.19	0.18	0.17	0.15	0.17	4.21

^aT9300 2.5 GHz.^bPentium 2.80 GHz.^cPentium 3 GHz.^dT7500 2.2 GHz.

SLC, PLA, and Reinforcement learning presented by Jędrzejowicz and Ratajczak-Ropel (2015, 2016) denoted respectively as JRPLA and JRR, hybrid particle swarm and differential evolution developed by Zhang et al. (2015) denoted as ZLZH, entropy-based upper bound methodology proposed by Chen et al. (2014) denoted as CLPE.

A comparison of average deviations can be made using Table 3. In this table the algorithms in descending performance based on average deviation have been ranked.

As it can be seen, the proposed GA has been located in top five algorithms in solving the MRCSP.

A comparison of CPU time must be drawn based on computing power (Chakraborty et al. 2016). Indeed, computers with powerful processors process a program in a short time (Vartouni and Khanli 2014). Comparing CPU time in Table 4 and considering lower computational power of this study compared to that of others, the CPU time of proposed algorithm can be considered satisfactory.

Conclusion

In this paper, a Genetic Algorithm (GA) was proposed to solve the MRCSP improved by a new local search method. An Activity Mode List (AML) was used as encoding and MSSGS was considered as the decoding procedure. In the feasible tackling procedure, a local search procedure was proposed in order to select activities with high potential for applying the mode improvement procedure on them. The local

search was applied in an example to demonstrate its capabilities to improve the makespan. Moreover, it was tested to solve the well-known benchmark set J20 in PSPLIB. The testing results were compared with the results of other existed local search and the proposed method proved its robustness in solving such problems. Also, the well-known benchmark set J20 in PSPLIB was used for testing the proposed algorithm. The performance comparisons indicated that the proposed GA is located in top five algorithms in solving the MRCSP. Furthermore, the results show that its computational time is satisfactory. The proposed model can be developed by future investigations in the directions below:

- Using fuzzy sets for duration of activities and comparing the findings.
- Extending the proposed method to solve various complex real world scheduling problems.

Disclosure statement

No potential conflict of interest was reported by the authors.

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